MISR UNIVERSITY
FOR SCIENCE & TECHNOLOGY
College of Information Technology



جامعة مصر
للعلوم والتكنولوجيا
كلية تكنولوجيا المعلومات

CS481- Pattern Recognition

Course Instructor

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“Education is the most powerful weapon which you can use to change the world.”

— Nelson Mandela

Expected Code of Conduct











Bayes Classifier

❑Probability

- ➢**Probability of an event is:**

$$\text{Probability of an event happening} = \frac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$$

Probability

- Each event has a probability value such that 0 <= P <= 1
- Summation of probabilities = 1
- Probability of Impossible Outcome = 0.0
- Probability of Certain Outcome = 1.0
- Probability of Not Event A = 1 – P(A)

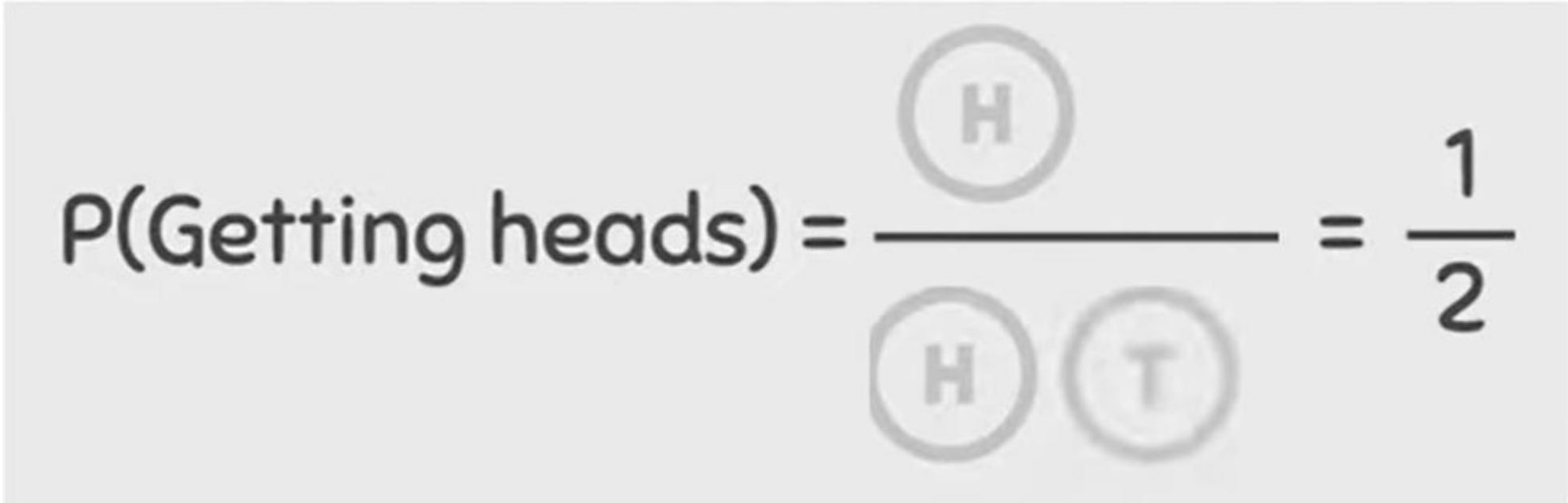
Probability

- If you roll a six-sided die, what is the probability of rolling a five?

$$P(\text{Rolling a 5}) = \frac{\text{one favorable outcome}}{\text{six possible outcomes}} = \frac{1}{6}$$

Probability

- If you flip a coin, what is the probability of getting a head?



Marginal probability

- **Marginal probability** is the probability of an event irrespective of another event.
- The probability of a specific value of one input variable is the marginal probability across the values of the other input variables.

		Smoker?		
		Yes	No	Total
Gender	Male	.41	.07	.48
	Female	.47	.05	.52
	Total	.88		1.00

- P(X=Male) = sum P(X=Male, Y=yi) for all i

Joint probability

- **Joint Probability** is the probability of **two events** occurring together and **at the same point in time.**
- The probability of having both events at once is simply refereed as P(A and B).
- Also P(A and B) can be denoted as $P(A \cap B)$ or $P(B \cap A)$.
- $P(A \text{ and } B) = P(A, B) = P(A \cap B) = P(B \cap A)$.

Joint probability

- If the occurrence of one does not affect the occurrence of the other, then these events are called **independent events**.
- Example: Rolling a die, and flipping a coin are independent events.
- Say the probability of getting a **head** is called **P(A)**. the probability of rolling **five** is called **P(B)**.



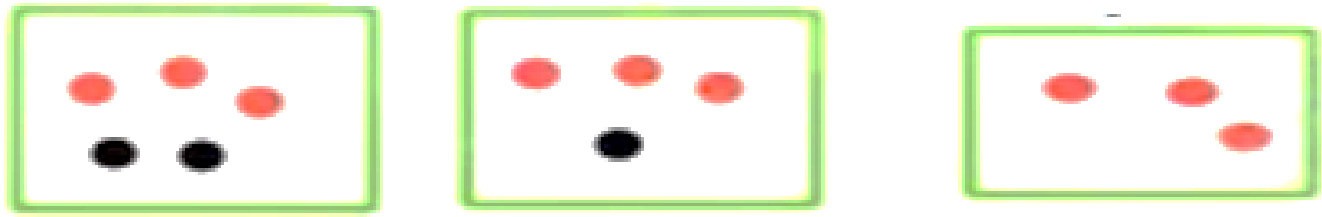
- **P(A and B) =P(A)*P(B).**

  P(B)=P(B|A) since B does not depend on A.

- Probability of getting a head and rolling five
- = P(getting a head)* P(rolling five)
- = (1/6) * (1/2)= 1/12= 0.0833

Joint probability

- If the occurrence of one affects the occurrence of the other, then these events are called **dependent events.**
- Assume we have a box with 3 red, and 2 black balls.
- If we pick a black ball (Event A), then pick another black ball from the rest of the balls (Event B).



- We say Event A, and Event B are dependent
- $P(A \cap B)=P(A)*P(B|A)$

$=(2/5)*(1/4)=1/10=0.1$

Joint probability

INDEPENDENT EVENTS

The outcome of one event does not affect the outcome of the other event.

$$P(A \cap B) = P(A) \times P(B)$$

DEPENDENT EVENTS

The outcome of one event does influence the outcome of the other event.

$$P(A \cap B) = P(A) \times P(B \text{ after } A)$$

Question:

In a class of 100 students, there are 60 boys and 40 girls. Among the boys, 40 are tall and 20 are short. Among the girls, 10 are tall and 30 are short.

- **Calculate the following:**

a) **What is the probability of randomly selecting a student who is both a girl and tall?**

Method 1: P(girl, tall)

Solution

- 100 Students (60 boys – 40 Girls)
  - Boys (60) (Tall 40 → Short 20)
  - Girls (40) (Tall 10 → Short 30)

- Probability calculations:

$$P(\text{girl, tall}) = P(\text{girl}) * P(\text{tall} \mid \text{girl})$$

$$= 40/100 * 10/40 = 10/100 = 10\%$$

$$P(\text{tall, girl}) = P(\text{tall}) * P(\text{girl} \mid \text{tall})$$

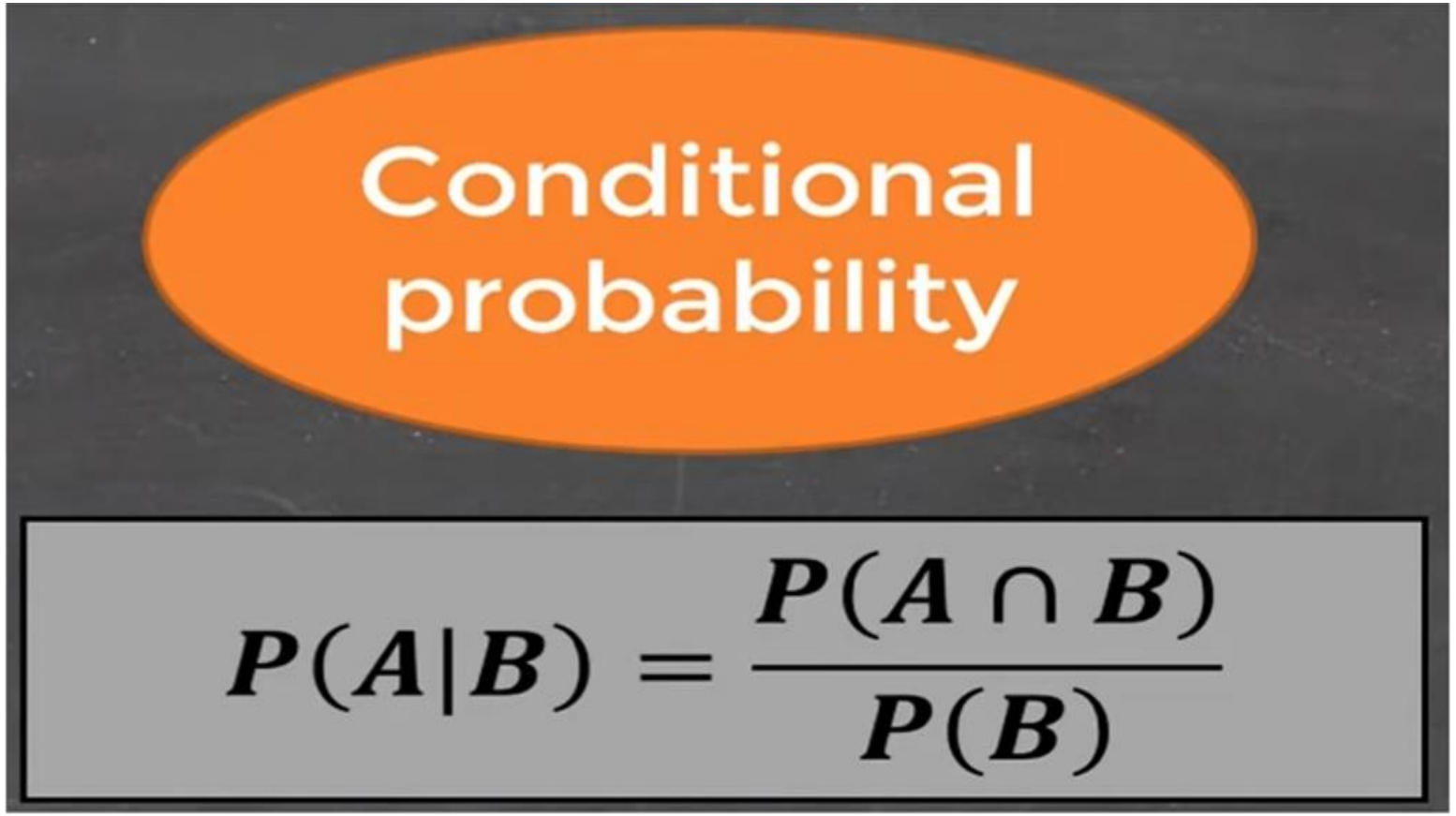
$$= 50/100 * 10/50 = 10\%$$

Conditional Probability

- The occurrence of two events, where they are dependent events is called **conditional probability**: P(A|B).
- Sometimes, many event occurs while one of them depends of the rest of events: P(B|A,C,D,E, F,..).
- **Bayes theorem** concerns with the dependency of an event on only one event: P(A|B)
- **Naive Bayes theorem** concerns with the dependency of an event on many events: P(A|B,C,D,E,F,...)

Conditional Probability

- Conditional probability equals joint probability over marginal probability.



Bayes Classifier

- Bayes' Theorem finds the probability of an event occurring given the probability of another event that has already occurred (conditional probability)
- Bayes Theorem is the conditional probability with one event.
- P(B|A): B changes with values of A only.
- Bayes' Theorem makes use of the fact that: $P(A \cap B) = P(B \cap A)$.

Bayes Classifier

- Since
  - $P(A \cap B) = P(B \cap A)$,
  - $P(A \cap B) = P(A) * P(B|A)$
  - $P(B \cap A) = P(B) * P(A|B)$

  Equal
- So,
  - $P(A)*P(B|A) = P(B)*P(A|B)$,
  - $P(A|B) = [P(A)*P(B|A)] / [P(B)]$
- We refer to
  - $P(A|B)$ as posterior probability,
  - $P(A)$ as prior probability
  - $P(B|A)$ as likelihood.
  - $P(B)$ as marginal probability

Bayes Classifier

- We call a probability a **prior probability** since it is the probability **available** before we observe the event. It is the complete information that we have first.
- We call a probability a **posterior probability** since it is the probability obtained after we have the event.

Example

Over the past 14 days, the weather conditions and whether Football player played outside were recorded as follows:

Part A: What is the probability that children will play outside given that the weather is overcast?

$$P(\text{Play} \mid \text{Overcast}) = ?$$

Part B: What is the probability that the weather is overcast given that Football player did NOT play outside?

$$P(\text{Overcast} \mid \text{No}) = ?$$

Weather	Play
Sunny	No
Sunny	No
Overcast	Yes
Rainy	Yes
Rainy	Yes
Rainy	No
Overcast	Yes
Sunny	No
Sunny	Yes
Rainy	Yes
Sunny	Yes
Overcast	Yes
Overcast	Yes
Rainy	No

Solution

P(Overcast | Play) =4/14

Weather	No Play	Yes Pla	Total
Overcast	0	4	4
Sunny	3	2	5
Rainy	2	3	5
Total	**5**	**9**	**14**

P(No) =9/14

P(Yes) =9/14

❑using Bayes' formula:

$$P(A|B) = P(A)] \times [P(B|A) / P(B)$$

❑**Part A:**

$$P(\text{Play} \mid \text{Overcast}) = P(\text{Play})] \times [P(\text{Overcast} \mid \text{Play}) / P(\text{Overcast})$$

$$= (9/14 \times 4/9) / (4/14) = 4/4 = 1$$

❑**Part B:**

$$P(\text{No} \mid \text{Overcast}) = [P(\text{Overcast} \mid \text{No}) \times P(\text{No})] / P(\text{Overcast})$$

$$= (0/5 \times 5/14) / (4/14) = 0$$

Example

Problem statement

- A new medical test is used to detect whether a patient has a certain cancer or not, whose test result is either + (*positive*) or - (*negative*)
- For patient with this cancer, the probability of returning a *positive* test result is 0.98 i.e $p(+|cancer)=0.98$
- For patient without this cancer, the probability of returning a *negative* test result is 0.97 i.e $p(-|nocancer)=0.97$
- The probability for any person to have this cancer is 0.008
  i.e $p(cancer)=0.008$

Question

If positive test result is returned for some person, does he/she have this kind of cancer or not? P(cancer|positive) or P(nocancer|positive) i.e $(p(w_1|+)$ *or* $p(w_2|+))$

Example 2

ω_1 : cancer $\quad$ ω_2 : no cancer $\quad$ $x \in \{+, -\}$

$$P(\omega_1) = 0.008 \qquad P(\omega_2) = 1 - P(\omega_1) = 0.992$$

$$P(+ \mid \omega_1) = 0.98 \qquad P(- \mid \omega_1) = 1 - P(+ \mid \omega_1) = 0.02$$

$$P(- \mid \omega_2) = 0.97 \qquad P(+ \mid \omega_2) = 1 - P(- \mid \omega_2) = 0.03$$

$$P(\omega_1 \mid +) = \frac{P(\omega_1)P(+ \mid \omega_1)}{P(+)} = \frac{P(\omega_1)P(+ \mid \omega_1)}{P(\omega_1)P(+ \mid \omega_1) + P(\omega_2)P(+ \mid \omega_2)}$$

$$= \frac{0.008 \times 0.98}{0.008 \times 0.98 + 0.992 \times 0.03} = 0.2085$$

$$P(\omega_2 \mid +) = 1 - P(\omega_1 \mid +) = 0.7915$$



Thank You